

1. Quiz: Change (Limits & Derivatives)

For each question, choose the best answer.

1.1. Question 1

The limit $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ equals:

- (a) 0 (b) 4 (c) Does not exist (d) Undefined at $x = 2$, so the limit doesn't exist either

1.2. Question 2

If $f'(x) > 0$ for all x in (a, b) and $f''(x) < 0$ for all x in (a, b) , then on this interval the function is:

- (a) Increasing and concave up (b) Increasing and concave down (c) Decreasing and concave up (d) Decreasing and concave down

1.3. Question 3

In a related rates problem, a ladder 5 m long leans against a wall. The bottom slides away at 2 m/s. How fast is the top sliding down when the bottom is 3 m from the wall?

- (a) 1.5 m/s (b) 2 m/s (c) 2.5 m/s (d) $\frac{3}{2}$ m/s

1.4. Question 4

For the function $f(x) = x^3 - 3x^2$, the point $x = 2$ is:

- (a) A local maximum (b) A local minimum (c) An inflection point (d) Neither a critical point nor an inflection point

1.5. Question 5

The derivative of $f(x) = \ln(\sin x)$ is:

- (a) $\frac{1}{\sin x}$ (b) $\cot x$ (c) $\frac{\cos x}{\sin x}$ (d) Both (b) and (c)

1.6. Question 6

To find the maximum of $f(x) = xe^{-x}$ on $[0, 2]$, you should:

- (a) Check $f(0)$, $f(2)$, and where $f'(x) = 0$ (b) Only check where $f'(x) = 0$ (c) Only check $f(0)$ and $f(2)$ (d) Check where $f''(x) = 0$

1.7. Question 7

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = 5$, then by L'Hôpital's rule:

- (a) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 5$ (b) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ does not exist (c) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$ (d) We cannot conclude anything without more information

1.8. Question 8

A function has a critical point at $x = c$. The second derivative test is inconclusive when:

- (a) $f''(c) = 0$ (b) $f''(c) > 0$ (c) $f''(c) < 0$ (d) $f''(c)$ does not exist

1.9. Question 9

For an optimization problem on a closed interval $[a, b]$, the absolute maximum:

- (a) Must occur at a critical point (b) Must occur at an endpoint (c) Must occur at either a critical point or an endpoint (d) Must occur where $f''(x) < 0$

1.10. Question 10

The equation of the tangent line to $y = x^2$ at $x = 3$ is:

(a) $y = 6x - 9$ (b) $y = 6x$ (c) $y = 9x - 18$ (d) $y = 2x + 3$